M.S. NAZ HIGH SCHOOL

Excellence in Matriculation & Academic Leadership | Est. 1989 | BISE Gujranwala Code: 112199

Single National Curriculum (SNC) & PECTAA Standards Aligned - Session 2026-2027

CLASS 9 (9TH MATRIC) - COMPUTER SCIENCE

Official Syllabus, Board Paper Scheme & High-Yield Examination Revision Notes

Group: Computer Science Group | Total Marks: 75 Marks (Theory: 50 | Practical: 25) | Time: 2 Hours 30 Minutes

SECTION 1: BISE GUJRANWALA EXAMINATION BLUEPRINT & WEIGHTAGE

- **Section A: Objective (MCQs):**
  10 Compulsory MCQs (1 Mark each = 10 Marks).
- **Section B: Short Questions:**
  Attempt 12 out of 18 Short Questions (3 sub-sections: Q2, Q3, Q4: 4/6 each = 24 Marks).
- **Section C: Long / Analytical Questions:**
  Attempt 2 out of 3 Long Questions (Q5, Q6, Q7: 8 Marks each = 16 Marks).
- **SLO Cognitive Weightage:**
  Algorithmic Problem Solving: 35% | Binary Logic & Security: 35% | Web Designing (HTML): 30%.

SECTION 2: CHAPTER-WISE HIGH-YIELD REVISION NOTES & CORE SLOs

Unit 1: Problem Solving:

Steps in Problem Solving (Defining, Understanding 5 Ws, Planning, Defining candid solutions, Selecting best solution), Flowcharts (Symbols: Terminal, Process, Decision, Input/Output, Connector, Flow lines), Importance of flowcharts, Algorithms (Formulation, Notation: Start, Input, Set, If-else, Output, Stop), Algorithm efficiency (Time & Space complexity), Verification vs Validation, Trace table and finding runtime errors.

Unit 2: Binary Number Systems:

Number Systems: Decimal (Base 10), Binary (Base 2), Hexadecimal (Base 16), Conversions (Decimal to Binary/Hex, Binary/Hex to Decimal, Binary to Hex grouping of 4 bits), Memory and Data Storage: Bit, Nibble, Byte, KB, MB, GB, TB, PB, Volatile (RAM) vs Non-Volatile (ROM) storage, Data Representation: Text (ASCII code, Unicode), Boolean Algebra (Propositions, Truth values, Logical Operators: AND, OR, NOT, Truth tables, Laws of Boolean Algebra).

Unit 3: Computer Networks:

Definition of Computer Network, Need for a network (file sharing, hardware sharing, internet sharing), Client-Server vs Peer-to-Peer architecture, Network Topologies (Bus, Star, Ring, Mesh - diagrams, pros & cons), Transmission Media: Guided (Twisted pair, Coaxial, Fiber optic) and Unguided (Radio waves, Microwaves, Infrared), Communication over Internet: TCP/IP Model (Application, Transport, Network, Data Link, Physical layer), IP Addressing (IPv4 32-bit vs IPv6 128-bit), Routers, DHCP, DNS.

Unit 4: Data and Cyber Security:

Ethical issues of security: Confidentiality, Integrity, Availability (CIA Triad), Piracy (Copyright, Patents, Trade Secrets), Cybercrimes: Identity theft, Phishing, DoS (Denial of Service) attacks, Hacking vs Cracking, Malware: Viruses, Worms, Trojan Horses, Spyware, Ransomware, Encryption: Plaintext, Ciphertext, Caesar cipher, Vigenere cipher, Strong passwords and 2FA.

Unit 5: Designing Website (HTML):

Introduction to HyperText Markup Language (HTML), Web browser vs Web server, URL structure, HTML tags (Paired vs Unpaired/Self-closing tags), Structure of HTML page (html, head, title, body), Text formatting tags (b, i, u, p, br, hr, h1 to h6), Lists in HTML: Ordered (ol), Unordered (ul), Definition (dl), Hyperlinks (a href), Images (img src, alt, width, height), HTML Tables (table, tr, th, td, colspan, rowspan).

SECTION 3: MSNS SENIOR FACULTY BOARD EXAMINATION STRATEGY

*** Practice drawing flowcharts with a ruler and correct diamond decision and oval terminal shapes.**

*** Master binary-hex conversions and ASCII bit calculation problems.**

*** Memorize HTML tag syntax with exact closing tags and quote-enclosed attributes.**